

PART 1: True/False Questions (2 points $\times$ 20 = 40 points).

Question	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Answer	F	F	T	F	F	T	T	F	F	T
Question	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
Answer	F	F	F	T	T	T	F	F	F	F

Q1. For a polytropic ideal gas, the specific internal energy difference between two states over an isobaric process ($P = \text{const.}$) is given by $\Delta u = c_p(T_2 - T_1)$.

Q2. A fluid can be condensed by isobaric cooling starting from a state defined by a pressure that is greater than the critical pressure.

Q3. The specific enthalpy at the outlet of a compressor is higher than that at its inlet.

Q4. The internal energy of an expanding gas contained in a non-adiabatic cylinder-piston always decreases.

Q5. The c_p of a fluid depends only on temperature.

Q6. A gas turbine engine can be more efficient than a Carnot engine if these engines do not operate between the same minimum and maximum temperatures.

Q7. A fluid kept at constant specific volume within a perfectly tight piston-cylinder system cannot transfer energy as work.

Q8. The internal energy of a macroscopic system is the sum of all types of microscopic energy modes and of macroscopic potential and kinetic energy.

Q9. While heating a saturated liquid at constant volume, its pressure and temperature remain constant.

Q10. A thermal energy reservoir (TER) can supply or absorb a finite amount of energy as heat without undergoing any change in temperature.

Q11. A thermodynamic engine can function without discharging energy as heat.

Q12. The coefficient of performance of a system can be obtained by dividing the useful energy output by the total energy input.

Q13. The specifications sheet of a car engine shows that its net efficiency is 45% at an ambient temperature of 23 °C. Your friend Jeremy tells you that the peak temperature of the gases in the cylinders is 230 °C. Is this true or false?

Q14. The internal energy of a gas is related to the energy associated with the microscopically disorganized motion of its molecules.

Q15. An isolated control volume containing an ideal gas undergoes an isothermal process. No internal energy gets accumulated within the control volume.

Q16. A person in a raising elevator lifting a box over a distance of half meter will do more work than a person on the ground lifting the same box over the same distance.

Q17. The terms of the equation for the steady-state energy balance of an open system are the same as those for a closed system.

Q18. The critical point can be evaluated using the ideal gas model.

Q19. In a polytropic process the pressure can change independently of the volume.

Q20. A system formed by a simple compressible substance enclosed in a constant volume can transfer energy as work and heat.